FRDM-MCXA156

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REVISION HISTORY

REVISION	DESCRIPTION OF CHANGE	DATE	Approved
X1	Draft	1-Jun-2023	Jonson Chen
A	Release	2-Jun-2023	Jonson Chen
A2	1. Populate R55 R57. Add Test Point. 2. Remove Cap and Resistors(C57 C58 R81 R82) Between USB Receptacle Shell and PCB Groud	12-Jan-2024	Jonson Chen
A3	Fix P2_6/LPSPI1_PCS2-ARD_D10 to P2_6/LPSPI1_PCS1-ARD_D10	8-April-2024	Jonson Chen
A4	Update MCU symbol	5-May-2024	Jonson Chen

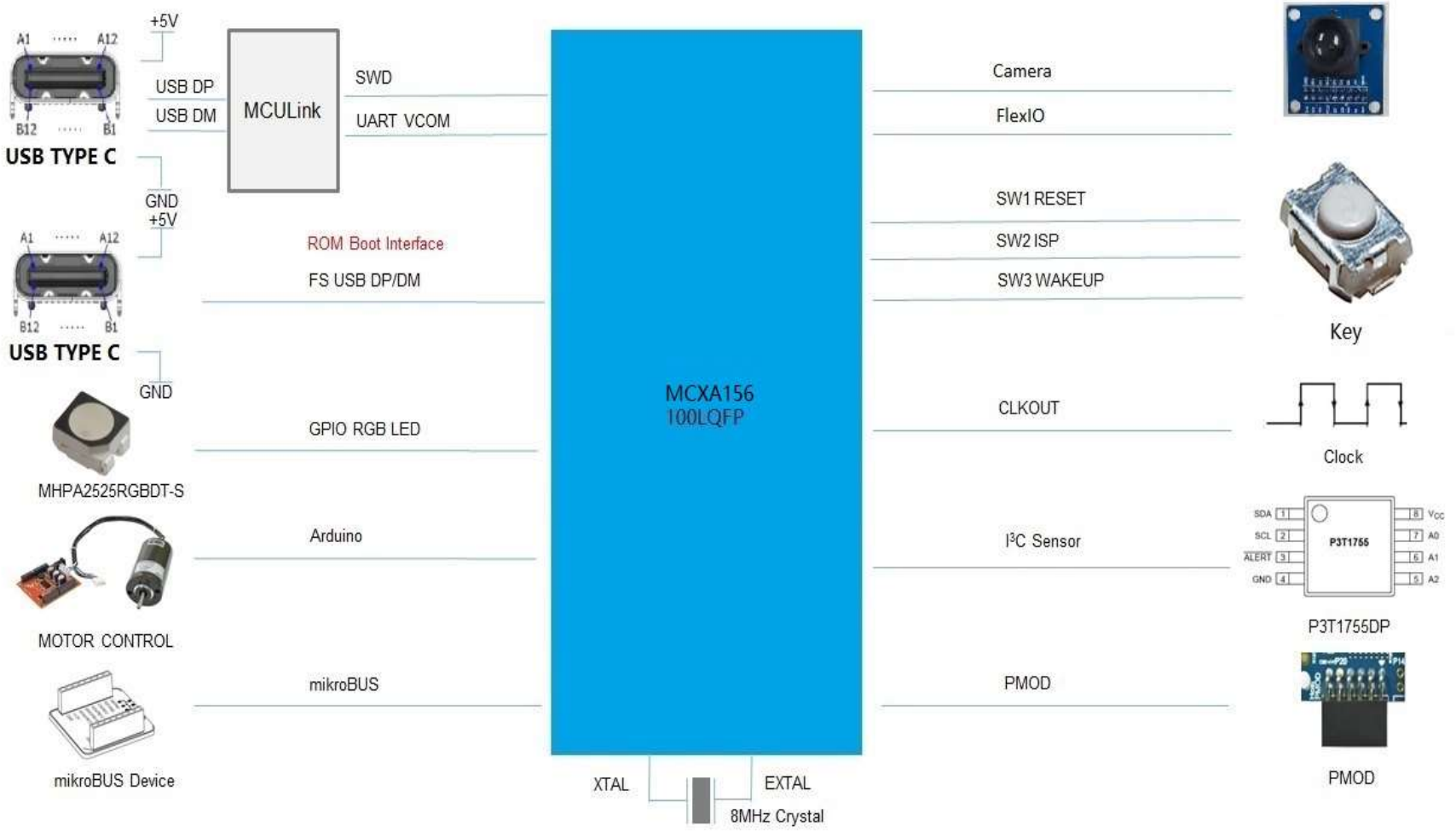
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	Date: Friday, July 19, 2024	Sheet 1 of 10

FRDM-MCXA156 BLOCK DIAGRAM



Classification: Company Internal/Proprietary		
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6	P0_0/SWDIO	TMS/SWDIO	76	P0_0/TMS/SWDIO/LPUART0_RTS_B/LPSPI0_PCS0/CT_INP0/FLEXIO0_D0
6	P0_1/SWCLK	TCLK/SWCLK	77	P0_1/TCLK/SWCLK/LPUART0_CTS_B/LPSPI0_SDI/CT_INP1/FLEXIO0_D1
3,6	P0_2	LPUART0_RXD/SWO	78	P0_2/UART_RXD/TDO/SWO/LPUART0_RXD/LPSPI0_SCK/CT0_MAT0/UTICK_CAP0/FLEXIO0_D2/I3C0_PUR
3,6	P0_3	LPUART0_TXD/TDI	79	P0_3/UART_TXD/CMP1_IN1/ADC0_A14/TDI/LPUART0_TXD/LPSPI0_SDO/CT0_MAT1/UTICK_CAP1/FLEXIO0_D3/CMP0_OUT
3,6,7	P0_6	CLKOUT	80	P0_6/ISPMODE_N/ADC0_A15/LPI2C0_HREQ/LPSPI0_PCS1/CT_INP2/FLEXIO0_D6/CMP1_OUT/CLKOUT/ADC1_CONV_CLK
3,7,8	P0_16	I3C0_SDA	83	P0_16/NVM_TM0/LPI2C0_SDA/LPSPI0_PCS2/CT0_MAT0/UTICK_CAP2/FLEXIO0_D0/I3C0_SDA/ADC1_CONV_DATA0/WUU0_IN2
3,7,8	P0_17	I3C0_SCL	84	P0_17/NVM_TM1/LPI2C0_SCL/LPSPI0_PCS3/CT0_MAT1/UTICK_CAP3/FLEXIO0_D1/I3C0_SCL/ADC1_CONV_DATA1
8	P0_18/FXIO_D2	FXIO_D2	85	P0_18/ADC0_A8/LPI2C0_SCLS/CT0_MAT2/FLEXIO0_D2/CMP0_OUT
8	P0_19/FXIO_D3	FXIO_D3	86	P0_19/ADC0_A9/LPI2C0_SDAS/CT0_MAT3/FLEXIO0_D3/CMP1_OUT
8	P0_20/FXIO_D4	FXIO_D4	87	P0_20/ADC0_A10/LPUART0_RXD/CT_INP0/FLEXIO0_D4
8	P0_21/FXIO_D5	FXIO_D5	88	P0_21/ADC0_A11/LPUART0_TXD/CT_INP1/FLEXIO0_D5
8	P0_22/FXIO_D6	FXIO_D6	89	P0_22/ADC0_A12/LPUART0_RTS_B/CT_INP2/CT0_MAT0/FLEXIO0_D6
8	P0_23/FXIO_D7	FXIO_D7	90	P0_23/ADC0_A13/LPUART0_CTS_B/CT_INP3/CT0_MAT1/FLEXIO0_D7
3,8	P1_0	LPSPI0_SDO	91	P1_0/SPI_SDO/ADC0_A16/CMP0_IN3/TRIG_IN0/LPSPI0_SDO/LPI2C1_SDA/CT_INP4/CT0_MAT2/FLEXIO0_D8/TP_DATA0/WUU0_IN6/LPTMR0_ALT3
3,8	P1_1	LPSPI0_SCK	92	P1_1/SPI_SCK/ADC0_A17/CMP1_IN3/TRIG_IN1/LPSPI0_SCK/LPI2C1_SCL/CT_INP5/CT0_MAT3/FLEXIO0_D9/TP_DATA1
3,8	P1_2	LPSPI0_SDI	93	P1_2/SPI_SDI/ADC0_A18/TRIG_OUT0/LPSPI0_SDI/LPI2C1_SDAS/CT1_MAT0/CT_INP0/FLEXIO0_D10/CAN0_TXD/USB0_OBE_TXDATA_b_TEST/TP_DATA2
3,8	P1_3	LPSPI0_PCS0	94	P1_3/SPI_PCS/ADC0_A19/CMP0_IN1/TRIG_OUT1/LPSPI0_PCS0/LPI2C1_SCLS/CT1_MAT1/CT_INP1/FLEXIO0_D11/CAN0_RXD/TP_DATA3/WUU0_IN7
8	P1_4/ADC0_A20-MC_BEMF_A	ADC0_A20	97	P1_4/ADC0_A20/CMP0_IN2/FREQME_CLK_IN0/LPSPI0_PCS3/LPUART2_RXD/CT1_MAT2/FLEXIO0_D12/USB0_IND_RXDM_TEST/ADC1_CONV_DATA2/WUU0_IN8
8	P1_5/ADC0_A21-MC_BEMF_B	ADC0_A21	98	P1_5/ADC0_A21/CMP1_IN2/FREQME_CLK_IN1/LPSPI0_PCS2/LPUART2_TXD/CT1_MAT3/FLEXIO0_D13/USB0_IND_RXDP_TEST/ADC1_CONV_DATA3
8	P1_6/TRIG_IN2-MC_ENC_I	TRIG_IN2	99	P1_6/ADC0_A22/NVM_TM2/TRIG_IN2/LPSPI0_PCS1/LPUART2_RTS_B/CT_INP6/CT4_MAT0/FLEXIO0_D14/CAN0_TXD/USB0_IND_RXDF_TEST/ADC1_CONV_DATA4
7	P1_7/WUU0_IN9-SW	WUU0_IN9	100	P1_7/ADC0_A23/NVM_TM3/TRIG_OUT2/LPUART2_CTS_B/CT_INP7/CT4_MAT1/FLEXIO0_D15/CAN0_RXD/USB0_DO_TXDATA_TEST/ADC1_CONV_DATA5/WUU0_IN9
3,8	P1_8	I2C2_SDA	1	P1_8/I2C_SDA/FREQME_CLK_IN0/LPUART1_RXD/LPI2C2_SDA/CT_INP8/CT0_MAT2/FLEXIO0_D16/I3C0_SDA/TP_HTRANS/WUU0_IN10
3,8	P1_9	I2C2_SCL	2	P1_9/I2C_SCL/FREQME_CLK_IN1/LPUART1_TXD/LPI2C2_SCL/CT_INP9/CT0_MAT3/FLEXIO0_D17/I3C0_SCL/TP_HWRITE
3,8	P1_10	ADC1_A8	3	P1_10/ADC1_A8/LPUART1_RTS_B/LPI2C2_SDAS/CT2_MAT0/FLEXIO0_D18/CAN0_TXD/USB0_PUE_PULLDWN_DP_TEST/ADC1_CONV_DATA6
7	P1_11/I3C0_PUR-P3T	I3C0_PUR	4	P1_11/ADC1_A9/TRIG_OUT2/LPUART1_CTS_B/LPI2C2_SCLS/CT2_MAT1/FLEXIO0_D19/I3C0_PUR/CAN0_RXD/USB0_OBE_TXBIAS_b_TEST/ADC1_CONV_DATA7/WUU0_IN11
3,8	P1_12	CAN_RXD	5	P1_12/ADC1_A10/LPI2C1_SDA/LPUART2_RXD/CT2_MAT2/FLEXIO0_D20/CAN0_RXD/USB0_DO_TXSZERO_TEST/WUU0_IN12
3,8	P1_13	CAN_TXD	6	P1_13/ADC1_A11/TRIG_IN3/LPI2C1_SCL/LPUART2_TXD/CT2_MAT3/FLEXIO0_D21/CAN0_TXD/USB0_SRE_LOWSPEED_TEST
3,8	P1_14	GPIO	7	P1_14/ADC1_A12/LPI2C1_SCLS/LPUART2_RTS_B/CT_INP10/CT3_MAT0/FLEXIO0_D22
3,8	P1_15	GPIO	8	P1_15/ADC1_A13/LPI2C1_SDAS/LPUART2_CTS_B/CT_INP11/CT3_MAT1/FLEXIO0_D23
6,7,8	P1_29/MCU_RESET_B	nRESET	9	P1_29/RESET_B/SPC_LPREQ/RESET_B
7	P1_30/XTAL48M	XTAL48M	10	P1_30/XTAL48M/TRIG_OUT3/LPI2C0_SDA/CT_INP16/FLEXIO0_D30/I3C0_SDA
7	P1_31/EXTAL48M	EXTAL48M	11	P1_31/EXTAL48M/TRIG_IN4/LPI2C0_SCL/CT_INP17/FLEXIO0_D31/I3C0_SCL
8	P2_0/FXIO_D8	ADC0_A0	22	P2_0/ADC0_A0/TRIG_IN6/LPUART0_RXD/LPUART4_CTS_B/CT_INP16/CT2_MAT0/FLEXIO0_D8/WUU0_IN18
3,8	P2_1	ADC0_A1	23	P2_1/ADC0_A1/TRIG_IN7/LPUART0_TXD/LPUART4_RTS_B/CT_INP17/CT2_MAT1/FLEXIO0_D9
8	P2_2/ADC0_A4-MC_CURRENT_DCB	ADC0_A4	24	P2_2/ADC0_A4/CMP0_IN0/DAC0_OUT/TRIG_IN8/LPUART0_RTS_B/LPUART2_TXD/CT_INP12/CT2_MAT2/FLEXIO0_D10
3,8	P2_3	ADC1_A4	25	P2_3/CMP1_IN0/ADC1_A4/TRIG_IN7/LPUART0_CTS_B/LPUART2_RXD/CT_INP13/CT2_MAT3/FLEXIO0_D11/NVM_SDO/WUU0_IN19
3,8	P2_4	ADC1_A0	26	P2_4/ADC1_A0/LPUART2_CTS_B/CT_INP14/CT1_MAT0/FLEXIO0_D12
3,8	P2_5	ADC1_A1	27	P2_5/ADC1_A1/LPUART2_RTS_B/CT_INP15/CT1_MAT1/FLEXIO0_D13
3,8	P2_6	ADC1_A3	28	P2_6/ADC1_A3/TRIG_OUT4/LPSPI1_PCS1/LPUART4_RXD/CT_INP18/CT1_MAT2/FLEXIO0_D14
8	P2_7/TRIG_IN5-MC_ENC_B	TRIG_IN5	29	P2_7/TRIG_IN5/LPUART4_TXD/CT_INP19/CT1_MAT3/FLEXIO0_D15/TP_HREADY/VREFI/ADC0_A7/ADC1_A7
8	P2_10/LPUART2_TXD-ARD_D1_TX	UART2_RX	31	P2_10/TRIG_OUT5/LPUART2_TXD/CT3_MAT2/FLEXIO0_D18
8	P2_11/LPUART2_RXD-ARD_D0_RX	UART2_TX	32	P2_11/TRIG_IN4/LPUART2_RXD/CT3_MAT3/FLEXIO0_D19
3,4,8	P2_12	LPSPI1_SCK	34	P2_12/USB0_VBUS_DET/ADC0_A5/OPAMP0_INP0/USB0_VBUS_DET/LPSPI1_SCK/LPUART1_RXD/CT4_MAT0/CT0_MAT0/FLEXIO0_D20/CAN0_RXD/WUU0_IN20
8	P2_13/LPSPI1_SDO-ARD_D11	LPSPI1_SDO	35	P2_13/ADC1_A5/TRIG_IN8/LPSPI1_SDO/LPUART1_TXD/CT4_MAT1/CT0_MAT1/FLEXIO0_D21/CAN0_TXD/OPAMP6_INN
8	P2_15/FXIO_LCD_INT	LCD_INT	36	P2_15/OPAMP0_OUT/ADC0_A2/TRIG_OUT4/LPSPI1_SDI/LPUART1_RTS_B/CT4_MAT3/CT0_MAT2/FLEXIO0_D23
3,8	P2_16	LPSPI1_SDI	37	P2_16/ADC0_A6/LPSPI1_SDI/LPUART1_RTS_B/CT3_MAT0/CT0_MAT2/FLEXIO0_D24
3,8	P2_17	TRIG_IN9	38	P2_17/ADC1_A6/TRIG_IN9/LPSPI1_PCS0/LPUART1_CTS_B/CT3_MAT1/CT0_MAT3/FLEXIO0_D25
8	P2_19/FXIO_LCD_CS	GPIO	39	P2_19/ADC1_A2/TRIG_OUT5/CT3_MAT3/FLEXIO0_D27
3,8	P2_20	GPIO	40	P2_20/TRIG_IN8/LPSPI1_PCS2/CT2_MAT0/FLEXIO0_D28
8	P2_21/FXIO_LCD_TE	FXIO_D28	41	P2_21/TRIG_IN9/LPSPI1_PCS3/CT2_MAT1/FLEXIO0_D29
8	P2_23/FXIO_LCD_WR	FXIO_31	42	P2_23/TRIG_OUT5/CT2_MAT3/FLEXIO0_D31

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3,8	P3_0	RGB_BLUE	73	P3_0/TRIG_IN0/LPUART3_RXD/CT_INP16/PWM0_A0/FLEXIO0_D8/PWM1_X0/TP_ADDR3/WUU0_IN22
3,8	P3_1	GPIO	72	P3_1/TRIG_IN1/LPUART3_TXD/CT_INP17/PWM0_B0/FLEXIO0_D9/PWM1_X1/FREQME_CLK_OUT0/TP_ADDR2
3,8	P3_6	PWM0_A0	71	P3_6/CLKOUT/LPSPI1_PCS3/LPUART3_RTS_B/CT4_MAT2/PWM0_A0/FLEXIO0_D14/PWM1_A0/FREQME_CLK_OUT1/ADC0_CONV_DATA0
8	P3_7/PWM0_B0-MC_PWM_AB	PWM0_B0	70	P3_7/TRIG_IN2/LPSPI1_PCS2/LPUART3_CTS_B/CT4_MAT3/PWM0_B0/FLEXIO0_D15/PWM1_B0/ADC0_CONV_DATA1
8	P3_8/PWM0_A1-MC_PWM_BT	PWM0_A1	69	P3_8/TRIG_IN3/LPSPI1_SDO/LPUART1_RXD/CT_INP4/PWM0_A1/FLEXIO0_D16/CLKOUT/ADC0_CONV_CLK/WUU0_IN23
8	P3_9/PWM0_B1-MC_PWM_BB	PWM0_B1	68	P3_9/TRIG_IN4/LPSPI1_SDI/LPUART1_TXD/CT_INP5/PWM0_B1/FLEXIO0_D17/TP_LOAD_EN
8	P3_10/PWM0_A2-MC_PWM_CT	PWM0_A2	67	P3_10/TRIG_IN5/LPSPI1_SCK/LPUART1_RTS_B/CT1_MAT0/PWM0_A2/FLEXIO0_D18/TP_ADDR1
8	P3_11/PWM0_B2-MC_PWM_CB	PWM0_B2	66	P3_11/TRIG_IN6/LPSPI1_PCS0/LPUART1_CTS_B/CT1_MAT1/PWM0_B2/FLEXIO0_D19/USB0_IBE_RXD/IFF_b_TEST/TP_ADDR0/WUU0_IN24
3,8	P3_12	PWM1_B2	62	P3_12/LPUART2_RTS_B/LPUART3_TXD/CT1_MAT2/PWM0_X0/FLEXIO0_D20/PWM1_A2/USB0_PUE_PULLDWN_DM_TEST/ADC0_CONV_DATA2
3,8	P3_13	PWM1_A1	61	P3_13/LPUART2_CTS_B/LPUART3_RXD/CT1_MAT3/PWM0_X1/FLEXIO0_D21/PWM1_B2/USB0_PUE_PULLUP_DP_TEST/ADC0_CONV_DATA3
3,8	P3_14	PWM1_B1	60	P3_14/LPUART2_RXD/LPUART3_CTS_B/CT_INP6/PWM0_X2/FLEXIO0_D22/PWM1_A1/USB0_SRE_LOWSPEED_TEST/ADC0_CONV_DATA4/WUU0_IN25
3,8	P3_15	PWM1_A0	59	P3_15/LPUART2_TXD/LPUART3_RTS_B/CT_INP7/FLEXIO0_D23/PWM1_B1/USB0_PUIMPSEL_PULLUP_DP_TEST/ADC0_CONV_DATA5
3,8	P3_16	PWM1_B0	58	P3_16/LPUART4_RTS_B/CT_INP8/FLEXIO0_D24/PWM1_A0
3,8	P3_17	PWM0_X0	57	P3_17/LPUART4_CTS_B/CT_INP9/FLEXIO0_D25/PWM1_B0
3,8	P3_18	PWM0_X1	56	P3_18/LPUART4_RXD/CT2_MAT0/PWM0_X0/FLEXIO0_D26/PWM1_X0
3,8	P3_19	UART1_RX	55	P3_19/LPUART4_TXD/CT2_MAT1/PWM0_X1/FLEXIO0_D27/PWM1_X1
3,8	P3_20	UART1_TX	54	P3_20/TRIG_OUT0/LPI2C3_SDA/LPUART1_RXD/CT2_MAT2/PWM0_X2/FLEXIO0_D28
3,8	P3_21	GPIO	53	P3_21/TRIG_OUT1/LPI2C3_SCL/LPUART1_TXD/CT2_MAT3/FLEXIO0_D29
8	P3_22/FXIO_LCD_GPIO	I2C3_SCL	52	P3_22/LPUART1_RTS_B/CT_INP10/FLEXIO0_D30/PWM1_X2
3,8	P3_27	I2C3_SDA	51	P3_27/TRIG_OUT7/LPI2C3_SCL/LPUART4_TXD/CT_INP13/CT3_MAT1/FLEXIO0_D27/WUU0_IN30
3,8	P3_28	GPIO	50	P3_28/TRIG_IN11/LPI2C3_SDA/LPUART4_RXD/CT_INP12/CT3_MAT2/FLEXIO0_D28/WUU0_IN26
8	P3_29/GPIO-MIKROE	ADC1_A21	49	P3_29/ADC1_A22/LPI2C3_HREQ/CT_INP3/CT3_MAT3/FLEXIO0_D29/TP_HRESP/WUU0_IN27
8	P3_30/ADC1_A21-MIKROE_AN	TRIG_IN10	48	P3_30/ADC1_A21/TRIG_OUT6/LPI2C3_SCLS/LPUART4_RTS_B/CT0_MAT2/FLEXIO0_D30/ADC0_CONV_DATA6
3,8	P3_31			P3_31/ADC1_A20/TRIG_IN10/LPI2C3_SDAS/LPUART4_CTS_B/CT0_MAT3/FLEXIO0_D31/ADC0_CONV_DATA7/LPTMR0_ALT2

8	P4_2/FXIO_D10	FXIO_D10	16	P4_2/CLKOUT/LPI2C2_SDAS/LPUART3_RXD/CT4_MAT0/PWM0_A2/FLEXIO0_D10/WUU0_IN16
8	P4_3/FXIO_D11	FXIO_D11	17	P4_3/LPI2C2_SCL/LPUART4_TXD/CT4_MAT1/PWM0_B2/FLEXIO0_D11
8	P4_4/FXIO_D12	FXIO_D12	18	P4_4/LPI2C2_SDA/LPUART4_RXD/CT4_MAT2/PWM0_A1/FLEXIO0_D12/WUU0_IN17
8	P4_5/FXIO_D13	FXIO_D13	19	P4_5/TRIG_OUT3/LPI2C2_SCLS/LPUART3_TXD/CT4_MAT3/PWM0_B1/FLEXIO0_D13
3,8	P4_6	FXIO_D14	20	P4_6/TRIG_IN4/LPI2C2_HREQ/LPUART3_CTS_B/CT_INP6/PWM0_A0/FLEXIO0_D14
8	P4_7/FXIO_D15	FXIO_D15	21	P4_7/TRIG_IN5/LPUART3_RTS_B/CT_INP7/PWM0_B0/FLEXIO0_D15
9	USB0_DP	USB0_DP	45	USB0_DP/USB0_DP/WUU0_IN29
9	USB0_DM	USB0_DM	44	USB0_DM/USB0_DM/WUU0_IN28

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3,8 P1_8/SDA — P1_8/LPI2C2_SDA-FXIO_HDR 8 / P1_8/LPI2C2_SDA-CAM_HDR 8

3,8 P1_9/SCL — P1_9/LPI2C2_SCL-FXIO_HDR 8 / P1_9/LPI2C2_SCL-CAM_HDR 8

3,8 P1_10 — P1_10/ADC1_A8-MC_CURRENT_DCB 8 / P1_10/ADC1_A8-ARD_A0 8

3,8 P1_15 — P1_15/GPIO-ARD_D8 8 / P1_15/GPIO-CAM_RESET 8

3,8 P2_1 — P2_1/ADC0_A1-MC_VOLT_DCB 8 / P2_1/FXIO_D9 8

3,8 P2_3 — P2_3/ADC1_A4-MC_VOLT_DCB 8 / P2_3/ADC1_A4-ARD_A2 8

3,8 P2_4 — P2_4/ADC1_A0-MC_BEMF_A 8 / P2_4/ADC1_A0-ARD_A3 8

3,8 P2_5 — P2_5/ADC1_A1-MC_BEMF_B 8 / P2_5/ADC1_A1-ARD_A1 8

3,8 P2_6 — P2_6/LPSPI1_PCS1-ARD_D10 8 / P2_6/ADC1_A3-MC_BEMF_C 8

3,8 P1_12 — R75 0 — P1_12/CAN_RXD 9 / P1_12/ADC1_A10-ARD_A4 8

3,8 P1_13 — R76 0 — P1_13/CAN_TXD 9 / P1_13/ADC1_A11-ARD_A5 8

3,8 P1_14 — P1_14/GPIO-ARD_D7 8 / P1_14/GPIO-CAM_PWDN 8

3,4,8 P2_12 — P2_12/LPSPI1_SCK-ARD_D13 4,8 / P2_12/USB0_VBUS_DET 4,8

3,8 P2_16 — P2_16/LPSPI1_SDI-ARD_D12 8 / P2_16/ADC0_A6-MC_BEMF_C 8

3,8 P2_17 — P2_17/TRIG_IN9-MC_ENC_B 8 / P2_17/FXIO_LCD_DC 8

3,8 P2_20 — P2_20/TRIG_IN8-MC_ENC_A 8 / P2_20/FXIO_LCD_RD 8

3,8 P3_6 — R83 0 — P3_6/PWM0_A0-MC_PWM_AT 8 / P3_6/CLKOUT 8

3,8 P3_15 — P3_15/PWM1_B1-ARD_D11 8 / P3_15/PWM1_B1-MC_PWM_BB 8

3,8 P3_21 — P3_21/LPUART1_TXD-MIKROE 8 / P3_21/FXIO_D29-CAM_PCLK 8

3,6 P0_2 — P0_2/SWO 6 / P0_2/LPUART0_RXD-MCULINK_VCOM_RX 6

3,6 P0_3 — P0_3/LPUART0_TXD-MCULINK_VCOM_TX 6 / P0_3/TDI 6

3,8 P3_0 — R72 0 — P3_0/PWM0_A0-BLED 7 / P3_0/FXIO_LCD_RST 8

3,8 P3_1 — P3_1/GPIO-ARD_D2 8 / P3_1/GPIO-CAM_VSYNC 8

3,8 P3_12 — R73 0 — P3_12/PWM1_A2-RLED 7 / P3_12/PWM1-ARD_D3 8 / P3_12/PWM1_A2-MC_PWM_CT 8

3,8 P3_13 — R74 0 — P3_13/PWM1_B2-GLED 7 / P3_13/PWM1_B2-ARD_D10 8 / P3_13/PWM1_B2-MC_PWM_CB 8

3,8 P3_14 — P3_14/PWM1-ARD_D5 8 / P3_14/PWM1_A1-MC_PWM_BT 8

3,8 P3_16 — P3_16/PWM1_A0-MC_PWM_AT 8 / P3_16/PWM1-ARD_D6 8

3,8 P3_17 — P3_17/PWM1-ARD_D9 8 / P3_17/PWM1_B0-MC_PWM_AB 8

3,8 P3_18 — P3_18/PWM0_X0-MIKROE 8 / P3_18/GPIO-CAM_HREF 8

3,8 P3_19 — P3_19/GPIO-PMOD 8 / P3_19/GPIO-MIKROE 8

3,8 P3_20 — P3_20/LPUART1_RXD-MIKROE 8 / P3_20/GPIO-PMOD 8

3,8 P1_0 — P1_0/LPSPI0_SDO-MIKROE 8 / P1_0/LPSPI0_SDO-PMOD 8

3,8 P1_1 — P1_1/LPSPI0_SCK-MIKROE 8 / P1_1/LPSPI0_SCK-PMOD 8

3,8 P1_2 — P1_2/LPSPI0_SDI-MIKROE 8 / P1_2/LPSPI0_SDI-PMOD 8

3,8 P1_3 — P1_3/LPSPI0_PCS-MIKROE 8 / P1_3/LPSPI0_PCS-PMOD 8

3,6,7 P0_6 — R84 0 — P0_6/ISPMODE_N-SW 7 / P0_6/ISPMODE_N-DEBUG 6,7 / P0_6/CLKOUT 6,7

3,7,8 P0_16 — R87 0 — P0_16/LPI2C0_SDA-ARD_D18 8 / P0_16/FXIO_D0 7,8 / P0_16/I3C0_SDA-P3T 7,8

3,7,8 P0_17 — R88 0 — P0_17/LPI2C0_SCL-ARD_D19 8 / P0_17/FXIO_D1 7,8 / P0_17/I3C0_SCL-P3T 7,8

3,8 P3_27/SCL — P3_27/LPI2C3_SCL-MIKROE 8 / P3_27/LPI2C3_SCL-PMOD 8

3,8 P3_28/SDA — P3_28/LPI2C3_SDA-MIKROE 8 / P3_28/LPI2C3_SDA-PMOD 8

3,8 P3_31 — P3_31/GPIO-ARD_D4 8 / P3_31/TRIG_IN10-MC_ENC_I 8

3,8 P4_6 — P4_6/TRIG_IN4-MC_ENC_A 8 / P4_6/FXIO_D14 8

VDD_BOARD — R79 4.7K, R80 4.7K

3,8 P1_8 — P1_8/SDA 3,8

3,8 P1_9 — P1_9/SCL 3,8

VDD_BOARD — R85 4.7K, R86 4.7K

3,8 P3_28 — P3_28/SDA 3,8

3,8 P3_27 — P3_27/SCL 3,8

Classification: Company Internal/Proprietary

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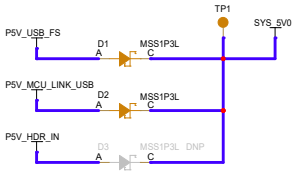
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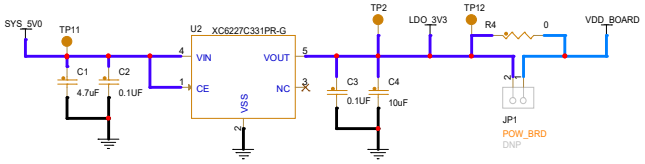
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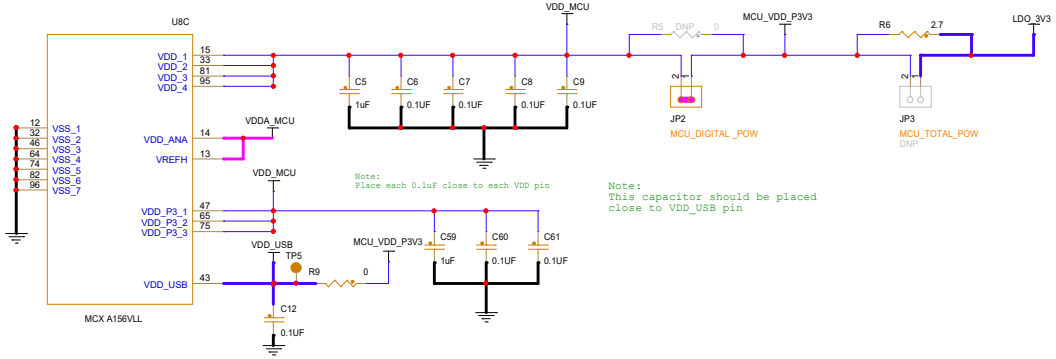
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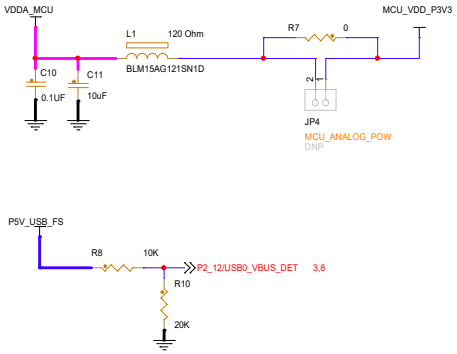
POWER

SYS_5V0

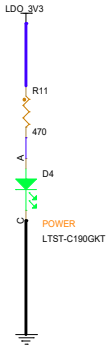




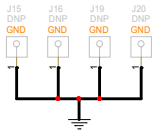




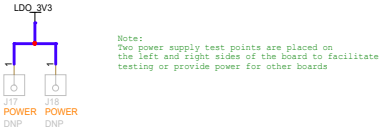
POWER LED



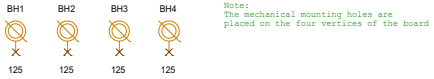
GND Test Points



Power Supply Test Points



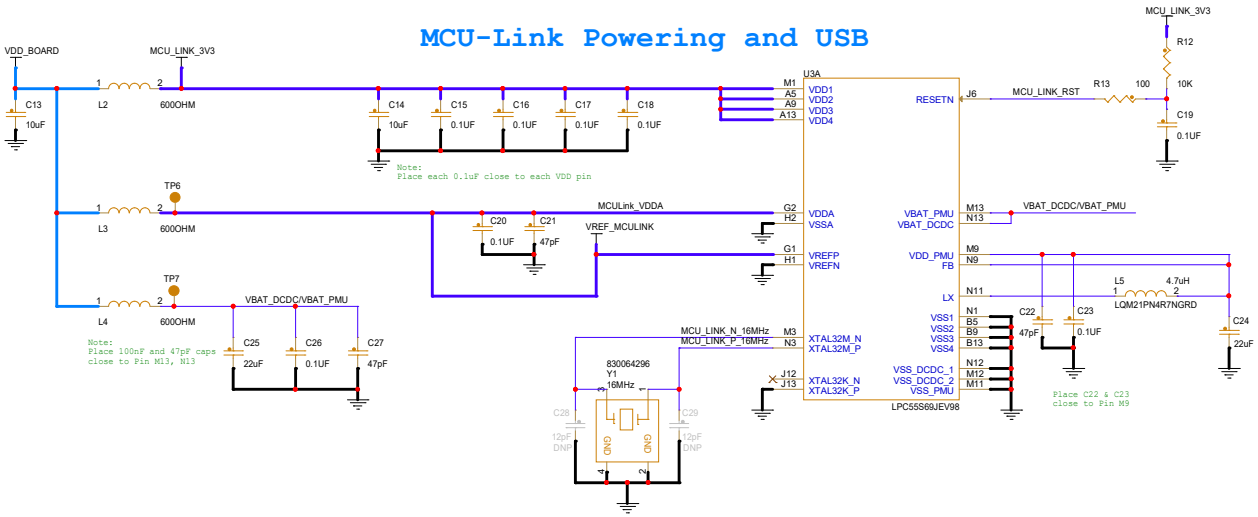
Mounting Holes



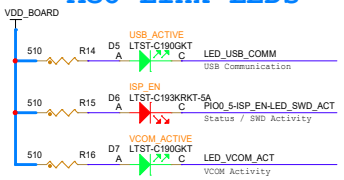
Classification: Company Internal/Proprietary

Drawing Title:	FRDM-MCXA156
Page Title:	POWER
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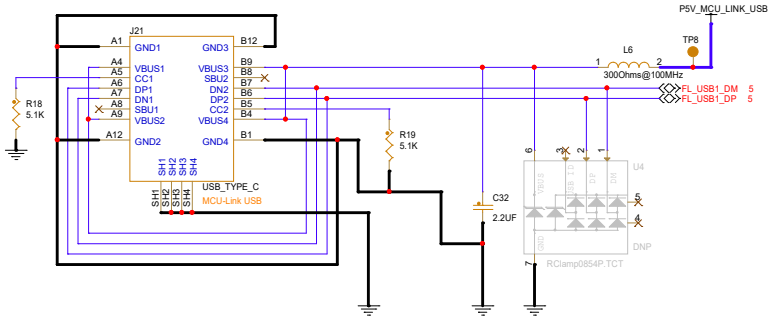
MCU-Link Powering and USB

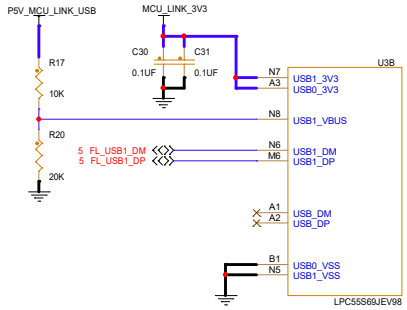


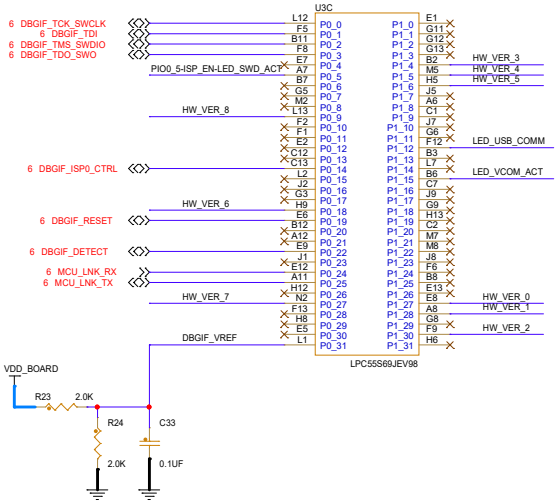
MCU-Link LEDs



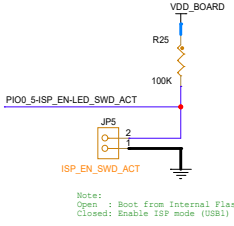
MCU-Link USB Type C

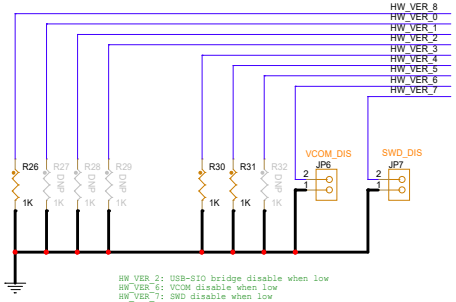






MCU-Link SWD connector





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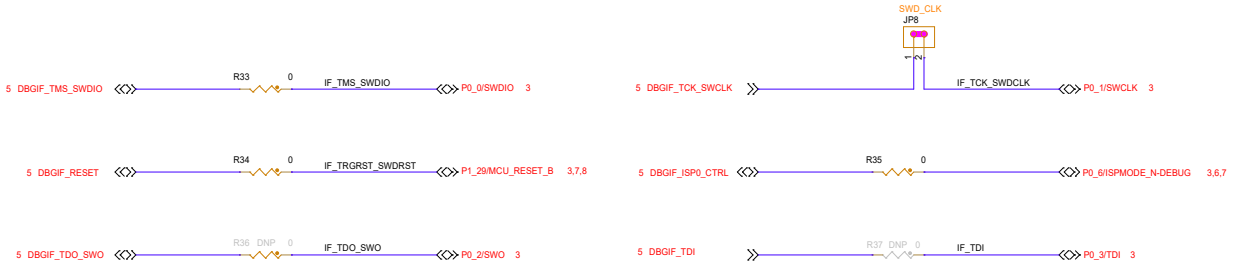
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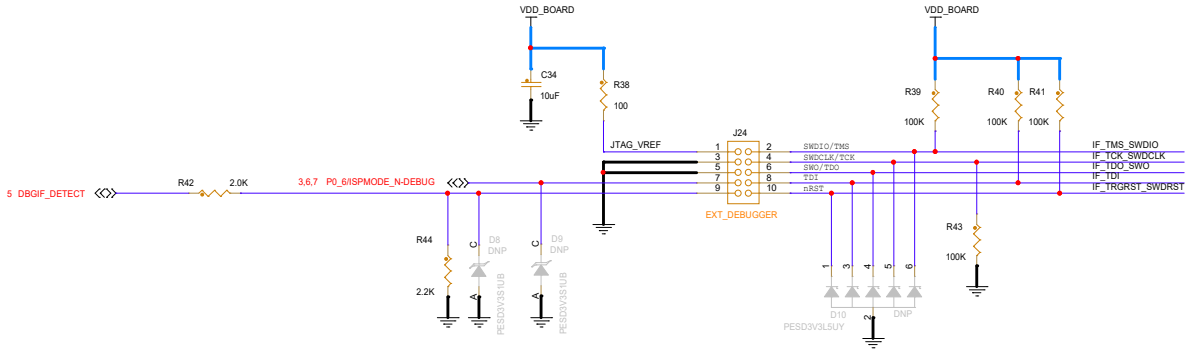
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MCU-Link Debug Interface



External Debugger



MCU VCOM



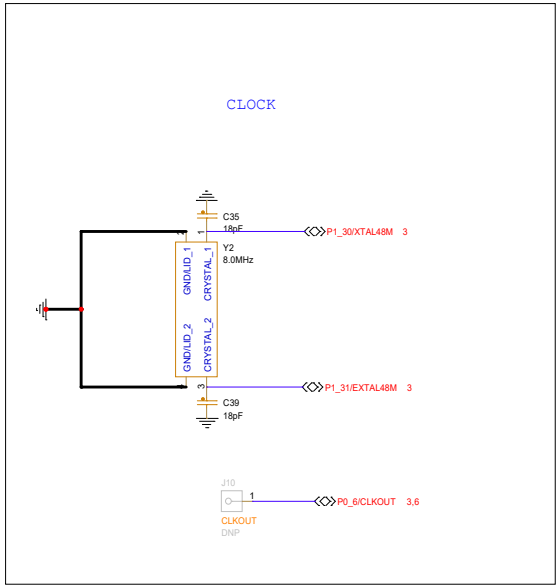
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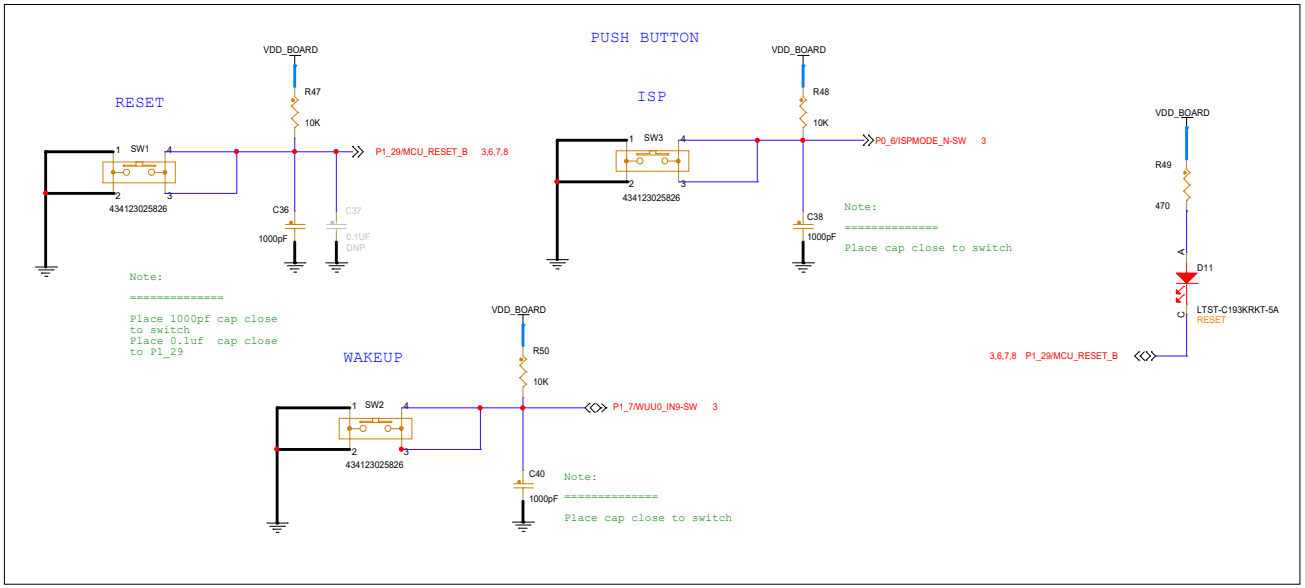
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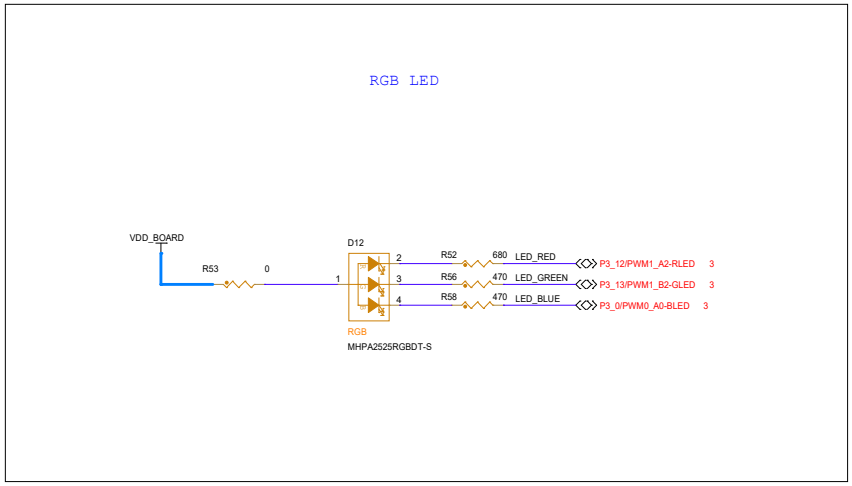
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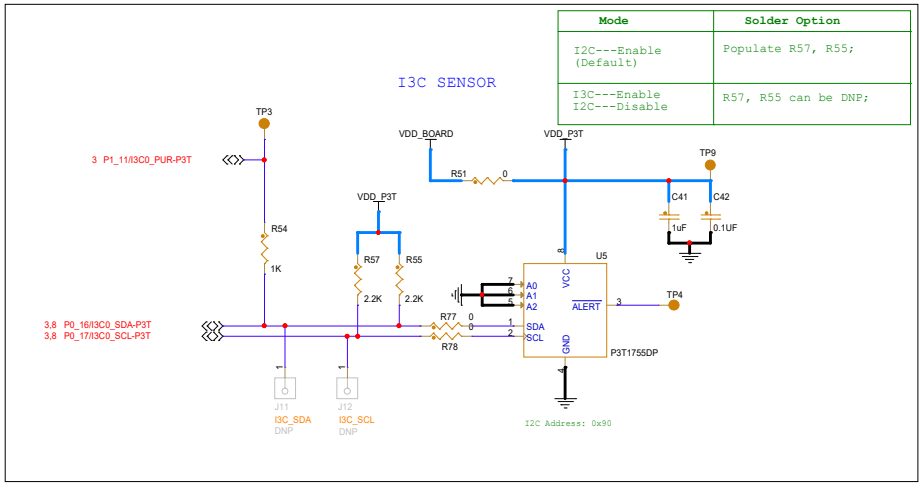
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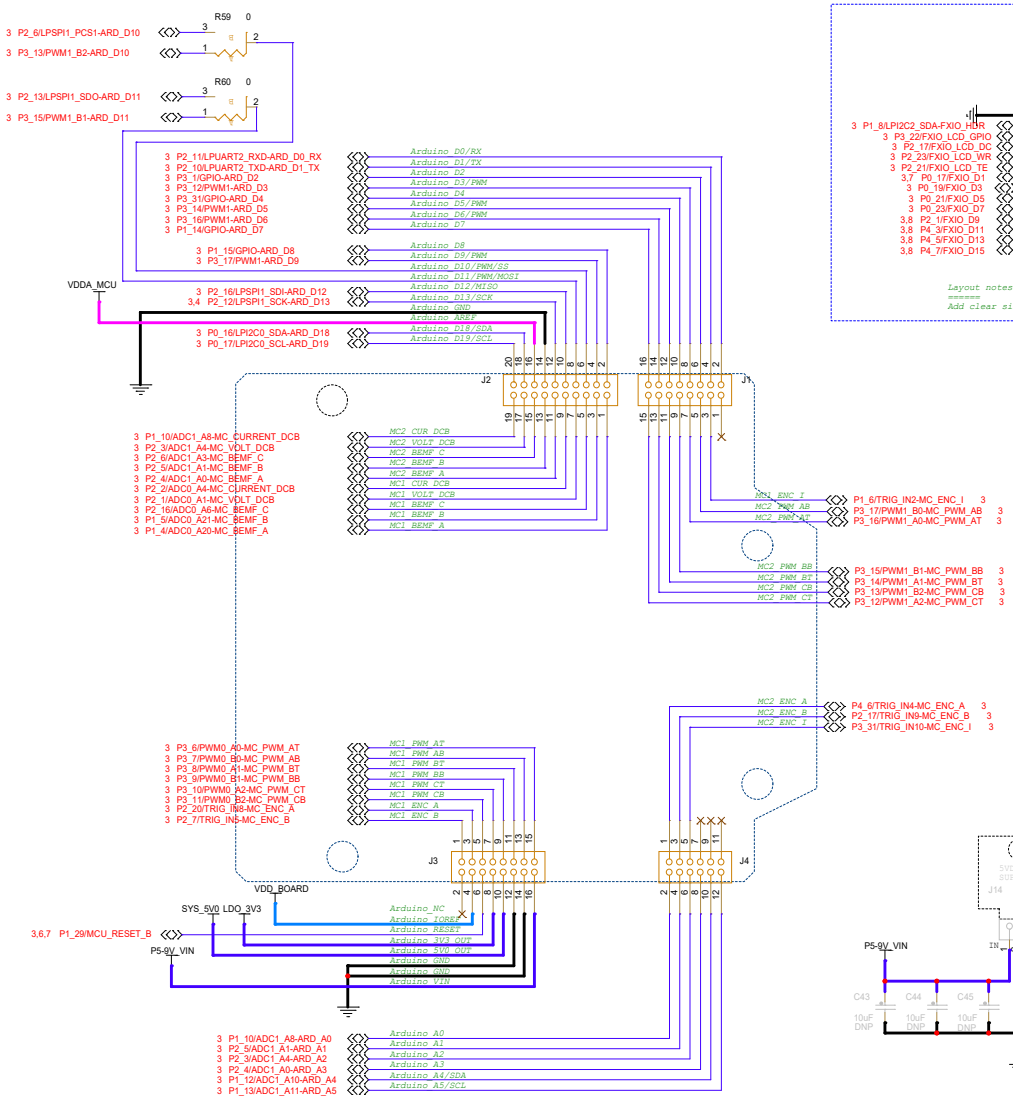
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Page Title: **CLOCK_RESET_HMI**

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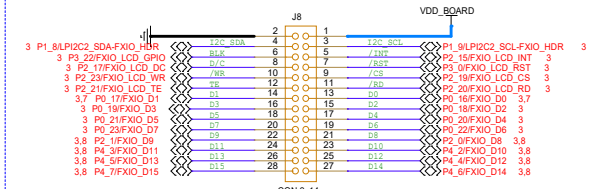
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ARDUINO SHIELD COMPATIBLE HEADERS



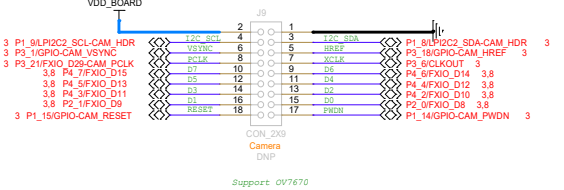
Shield Boards:
1. Arduino_: Arduino UNO R3
2. Motor control: FRDM_MC_LVBLDC and FRDM_MC_LVPMSM
3. Arduino A4/A5 used, R75,R76 should removed

FlexIO



Layout notes:
Add clear silkscreens for each pins of these connectors in this page.

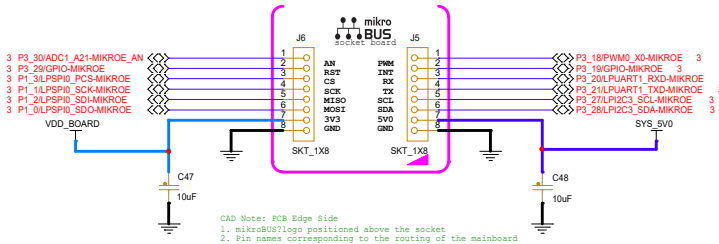
Camera



Support OV7670

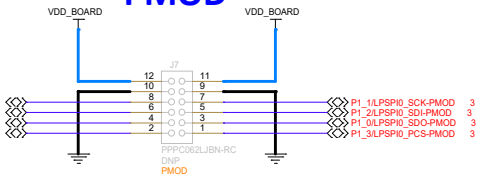
Layout notes:
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Mikro Bus



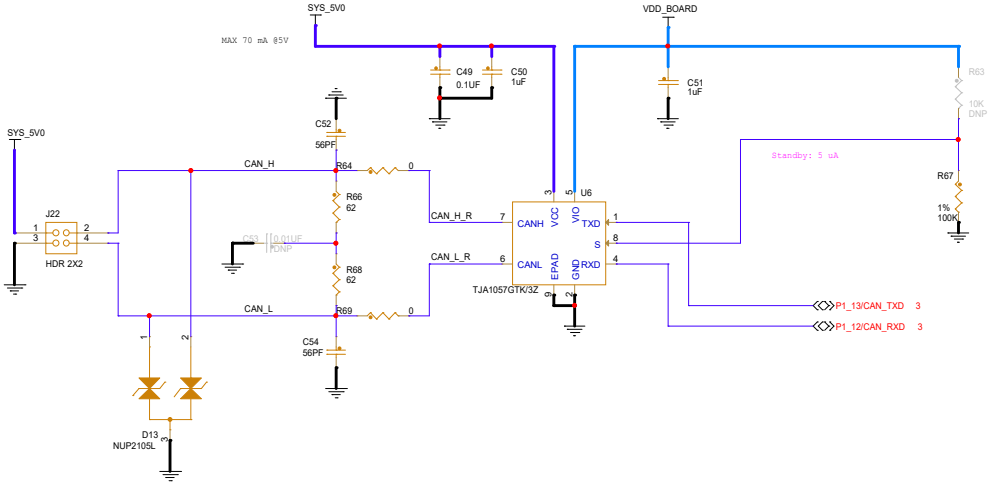
CAD Note: PCB Edge Side
1. mikroBUS?logo positioned above the socket
2. Pin names corresponding to the routing of the mainboard
3. A line that encloses the socket on all sides except the top.
4. A diagonal notch below the right-hand side pinout
(serves as a guideline for add-on board insertion)

PMOD



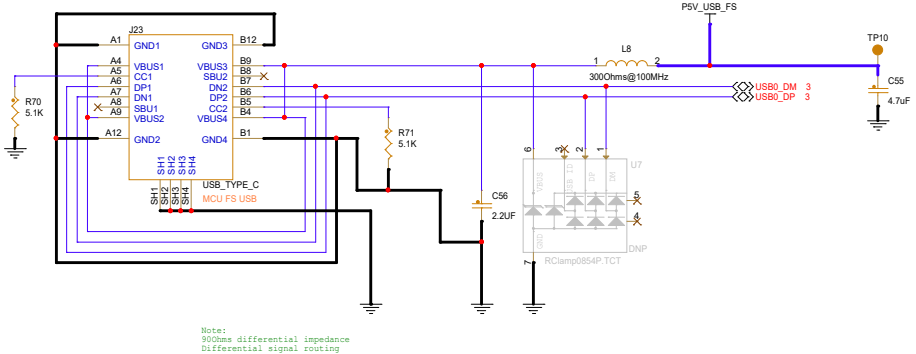
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Drawing Title: FRDM-MCXA156			
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FlexCAN



USB

USB FULL SPEED
USB Type C



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Page Title: **CAN_USB**

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REF DES	*JUMPER*	*PAGE NAME*
JP2	1-2	04 POWER
JP1,JP3,JP4	OPEN	04 POWER
JP5,JP6,JP7	OPEN	05 MCU_LINK_USB
JP8	1-2	06 MCU_LINK_DEBUG

REF DES	*ASSY_OPT*	*PAGE NAME*
D3,R5	DNP	04 POWER
JP1,JP3,JP4,J15,J16,J17,J18, J19,J20	DNP	04 POWER
C28,C29,R27,R28,R29,R32,U4	DNP	05 MCU_LINK_USB
D8,D9,D10,R36,R37	DNP	06 MCU_LINK_DEBUG
C37,J10,J11,J12	DNP	07 CLOCK_RESET_HMI
C43,C44,C45,C46,J7,J9,J14	DNP	08 HEADERS
U7,C53,R63	DNP	09 CAN_USB



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Drawing Title:

FRDM-MCXA156

Page Title:

APPENDIX JUMPER/DNP

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